Project 3: File Encryption Tool

In this project, we will create a **File Encryption Tool** that allows users to encrypt and decrypt text files using basic encryption algorithms. Encryption is an essential concept in the realm of data security, and understanding how it works can help safeguard sensitive information. This project will teach you the fundamentals of encryption and decryption processes, along with how to implement them using Python.

The File Encryption Tool will be able to:

- **Encrypt**: Convert a text file into an unreadable format to protect its contents.
- **Decrypt**: Convert the encrypted file back to its original form using a decryption key or method.

The project will focus on using basic encryption algorithms to demonstrate how file security works. We will start with a simple **Caesar Cipher**, which is one of the most basic encryption methods, and later, you can enhance it by exploring more advanced algorithms.

Step-by-Step Guide to Build the File Encryption Tool

Step 1: Understand the Encryption Algorithm

In this project, we will use the **Caesar Cipher** algorithm, which is a substitution cipher where each letter in the text is replaced by a letter some fixed number of positions down the alphabet. For example, with a shift of 3, the letter 'A' would become 'D', 'B' would become 'E', and so on.